

Investigation of Primitive-Space Scene Change Detection in Under-water 3D Gaussian Splatting Models

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Project Overview

1.1 Motivation and Background

Coral reefs are among the most biologically diverse and ecologically vital environments on the planet. However, they are increasingly vulnerable to climate change, pollution, and anthropogenic stressors. Effective biological research and environmental monitoring rely heavily on robust, high-frequency structural and health assessments. Scene Change Detection (SCD) has possible applications as a critical tool in these efforts, enabling researchers to track fine-scale structural dynamics (e.g., coral growth and physical breakage) as well as appearance shifts (e.g., coral bleaching) over time.

Recently, 3D Gaussian Splatting (3DGS) [1] has revolutionized 3D scene reconstruction. Its adaptation to underwater environments is a highly active but nascent field, with foundational methods only emerging between 2024 and 2026 [2–5]. These cutting-edge techniques successfully model and strip away complex underwater optical degradation, such as light scattering and attenuation. This project seeks to capitalize on these new methods, bridging the gap between novel 3D reconstruction research and practical, actionable tools for marine ecology.

1.2 Project Objectives

This project aims to investigate the feasibility and performance of a state-of-the-art primitive-space Scene Change Detection method [6] when applied to the challenging domain of underwater coral scenes. While this novel SCD method has demonstrated success on standard, terrestrial datasets, it is imperative to test its robustness in highly complex, real-world domains to determine its true utility and limitations.

Instead of relying on traditional render-then-compare paradigms that suffer from viewpoint dependencies and require pixel-space comparison, the project will perform

change detection directly on the native 3DGS primitives (analyzing position, anisotropic covariance, and color) [6]. To achieve this, the project will focus on two core evaluations:

1. **GS-DIFF Efficacy in Coral Environments:** Evaluate the success of the primitive-space change detection algorithm in accurately identifying and separating structural changes from appearance changes in underwater coral scenes.
2. **Evaluation of Underwater 3DGS Methods:** Research and select 2 to 4 of the most recent underwater 3DGS generation methods. The project will evaluate which of these techniques produces the most consistent and reliable reconstructions to serve as the optimal foundation for the change detection pipeline.

1.3 Scope and Context

To validate this pipeline, the project will utilize the EcoRRAP dataset [7], provided by the Australian Institute of Marine Science (AIMS) in collaboration with the QUT Centre for Robotics. Ultimately, this research will demonstrate how emerging underwater 3DGS rendering techniques can be harnessed to extract meaningful, temporal data, empowering biologists with advanced tools for reef monitoring.

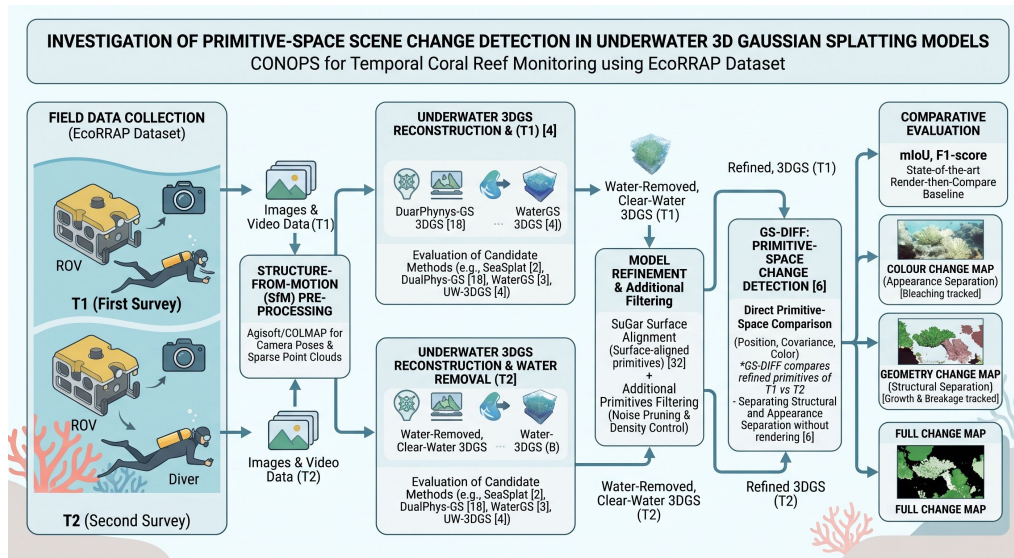


Figure 1.1: Concept of Operations (CONOPS) diagram outlining the proposed workflow and evaluation pipeline. *Note: The initial design of this diagram was created with the assistance of Gemini Nano Banana 2 and has been subsequently modified by the author.*

Literature Review

The literature review is structured into three main themes relevant to this project: the ecological necessity of coral reef monitoring, the evolution of scene change detection methods, and the recent rapid development of underwater 3D Gaussian splatting techniques.

2.1 The Importance of Coral Monitoring and Traditional Methods

Coral reef monitoring forms the fundamental baseline for marine conservation, providing the critical data necessary to detect long-term ecological trends, evaluate management interventions, and provide early warning systems for disturbances such as mass bleaching and disease outbreaks [8]. Traditional monitoring methodologies predominantly rely on diver-based in-situ surveys, utilizing techniques such as Manta Tows for broad-scale assessment, Point Intercept Transects (PIT) for benthic composition, and fixed-site photo-quadrats for localized structural tracking [9]. However, as highlighted by a recent comprehensive review of coral reef condition indicators by Davey et al. [10], there is a pervasive disconnect between the vast amounts of monitoring data collected globally and its actual application in spatial conservation planning. Traditional 2D metrics like percentage live coral cover, while ubiquitous, often fail to capture the complex, 3D structural dynamics necessary for holistic ecosystem assessment [10]. This limitation underscores the urgent need for advanced, scalable, and structurally aware monitoring technologies such as 3D Gaussian Splatting that can bridge the gap between data collection and actionable conservation insights.

2.2 Scene Change Detection and 3D Representations

Render-then-Compare Paradigms vs. Primitive-Level Comparison. Traditional scene change detection (SCD) across unconstrained viewpoints has historically operated in image space, relying on learned features or foundation models. Recent methods leverage 3D Gaussian Splatting (3DGS) [1] to synthesize reference views at novel poses, scoring changes by rendering views and comparing features or pixels [11]. While these render-then-compare approaches successfully suppress view-inconsistent distractors, they fundamentally use 3D primitives merely as a rendering substrate, relying on 2D space for the actual change signals. This often struggles to explicitly separate structural changes from mere illumination or appearance shifts.

In contrast, recent advancements have introduced a new paradigm operating directly within the 3D representation space. GS-DIFF [6] pioneered direct primitive-space comparison by analyzing the core attributes of 3D Gaussians (position, covariance, and color). The primary strength of this approach is its ability to separate structural transformations from superficial appearance shifts without relying on 2D image synthesis or external supervision. Similarly, 3DGS-CD [12] leverages 2D segmentation priors fused into the 3D space to detect physical object rearrangements rapidly, without requiring depth inputs or pre-defined object models. However, the critical limitation of primitive-level comparison is the inherently under-constrained nature of 3DGS; independent optimization runs can yield drastically different primitive configurations (Gaussian drift) even for unchanged scenes. This necessitates complex modeling of geometric and photometric drift to prevent false positives [6].

Spatial Consistency Constraints. To address the spatial inconsistency often found in localized change masks, methods like SCAR-3D [13] propose multi-view aggregation frameworks. By employing signed-distance-based differencing followed by voting and pruning mechanisms, this approach robustly separates pre- and post-change states. While highly accurate in producing consistent change masks, exploring non-rigid deformations within this framework remains an ongoing challenge.

2.3 Underwater Scene Reconstruction and 3D Gaussian Splatting

Underwater image formation is heavily degraded by absorption, scattering, and color shift, historically addressed via single-image enhancement [14] or Neural Radiance Fields [15]. With the advent of 3DGS, the literature has diverged into three primary methodological mindsets to tackle these aquatic challenges.

Approach 1: Physically-Grounded Image Formation (Inverse Rendering). Several works directly embed mathematical models of light attenuation and scattering into the splatting optimization pipeline. SeaSplat [2] constrains 3DGS using range- and color-dependent models, while WaterGS [3] formulates reconstruction as a formal inverse rendering problem. UW-3DGS [4] employs a learnable image-formation module paired with Physics-Aware Uncertainty Pruning, and DualPhys-GS [16] introduces dual feature-guided scattering-attenuation models. RecGS [17] specifically targets physical artifacts like water caustics. *Strengths & Limitations:* The core advantage of this physical mindset is its ability to disentangle true object color and geometry from the water medium, yielding highly accurate, structurally consistent primitives. However, this comes at a significant computational cost. Furthermore, these methods are highly sensitive to the accuracy of depth priors (which are notoriously difficult to obtain underwater) and often struggle in extreme long-distance scenes where the scattering signal overwhelmingly dominates the object signal.

Approach 2: Prior-Driven and Image Restoration Fusion. An alternative mindset eschews explicit physical equations in favor of leveraging pre-enhanced views or data-driven image restoration priors. Fast methods like WaterSplatting [18] and R-Splatting [19] bridge traditional 2D underwater restoration networks with the 3D structural constraints of Gaussian splatting. WaterClear-GS [20] uses optical-aware dual-branch optimization to clean textures. *Strengths & Limitations:* By optimizing for visual appearance rather than strict physics, these methods train significantly faster and frequently achieve superior perceptual quality metrics (e.g., PSNR, SSIM). Nevertheless, they are highly susceptible to view-inconsistencies introduced by 2D filters, are prone to overfitting localized noise, and inherently lack the capability to model distant volumetric media correctly since they rely entirely on the explicit geometric representations of 3DGS.

Approach 3: Multi-Modal Sonar Fusion. To overcome the severe visual degradation in turbid waters, a growing body of work integrates acoustic sensor data. SonarSplat [21] and NAS-GS [22] synthesize novel sonar views while modeling acoustic streaking phenomena. Crucially, Aqua-Splat [23] and SonarReg-GS [24] fuse Forward-Looking Sonar (FLS) measurements to serve as sparse metric depth anchors. *Strengths & Limitations:* Sonar effectively penetrates highly turbid water, and fusing it with 3DGS elegantly solves the monocular depth scale ambiguity problem that plagues purely visual underwater SLAM. Conversely, the limitations are severe: raw sonar observations are highly sparse, noisy, and suffer from elevation angle ambiguities. Consequently, while sonar fusion corrects global scale, it struggles to provide the dense geometric recovery required for fine-grained structural modeling without complex multi-sensor calibration.

Geometric Adaptations for Extreme Environments. Finally, to survive dynamic elements and low-visibility, some methods directly alter the structural mechanics of the Gaussians. Water-Adapted 3DGS [5] introduces complexity-adaptive point distributions, while RUSplatting [25] uses frame interpolation for sparse views. Dynamic distractors (e.g., marine life) are filtered by UW-GS [26] and Spatiotemporal 3DGS [27]. For wide-field observation, Underwater360 [28] leverages omnidirectional splatting. Controlled cross-regime studies [29] and surface-aligned mesh extractions like SuGaR [30] further validate the fidelity of these environmental adaptations.

2.4 Summary of Findings and Research Gap

The literature demonstrates a clear trajectory, the field has rapidly moved from adapting basic 3DGS to underwater scenes via standard image restoration priors, to embedding complex physical light-attenuation models directly into the optimization, and finally to adapting the structural primitives themselves to survive extreme turbidity and integrate multi-modal sensors. For coral reef ecosystems which present highly complex geometries, transient marine life, and dynamic lighting conditions these specialized underwater adaptations are paramount.

Table 2.1 provides a consolidated overview of the various 3DGS methods and their primary contributions to scene change detection and underwater reconstruction.

This project aims to bridge the critical gap between advanced underwater 3DGS reconstruction (Section 2.3) and primitive-space scene change detection (Section 2.2). By

Table 2.1: Summary of 3DGS Methods and Approaches

Category	Method	Key Approach / Contribution
Scene Change Detection	GS-DIFF [6]	Direct primitive-space comparison (position, covariance, color).
	3DGS-CD [12]	Detects physical object rearrangements using explicit splat geometric properties.
Physically-Grounded Underwater 3DGS	SeaSplat [2]	Physically grounded range- and color-dependent modeling.
	WaterGS [3]	Formulates scene reconstruction as an inverse rendering problem.
	DualPhys-GS [16]	Dual feature-guided scattering-attenuation modeling for selective wavelength attenuation.
	UW-3DGS [4]	Learnable image-formation with Physics-Aware Uncertainty Pruning.
	RecGS [17]	Targets and removes specific physical artifacts like water caustics.
Image Restoration & Prior-Driven	WaterClear-GS [20]	Optical-aware dual-branch optimization using restoration priors.
	Fast Methods [18, 19]	Bridges traditional 2D underwater restoration networks with 3D structural constraints.
Extreme Environments & Adaptations	Water-Adapted 3DGS [5]	Complexity-adaptive points and depth-adaptive multi-scale rendering.
	RUSplatting [25]	Frame interpolation and decoupled RGB learning for sparse-view/low-light.
	UW-GS [26]	Distractor-aware variance filtering for transient marine life.
	Spatiotemporal 3DGS [27]	Models localized turbidity and current variations.
	Underwater360 [28]	Omnidirectional splatting to eliminate refraction distortion.
Multi-Modal Sonar Fusion	SonarSplat [21], NAS-GS [22]	Synthesizes novel sonar views and handles noise-aware splatting.
	Aqua-Splat [23], SonarReg-GS [24]	Fuses sonar for sparse metric depth anchors to correct scale ambiguity.

evaluating which of the state-of-the-art underwater 3DGS methods produces the most geometrically robust and physically accurate primitives for coral scenes, this project will establish a foundational pipeline for tracking structural reef changes. Ultimately, this will move coral monitoring beyond simple 2D metrics (Section 2.1) and provide the rich, 3D structural data necessary to support effective spatial conservation planning.

Stakeholders & Resources

3.1 Stakeholders

The stakeholders for this project are identified in the table below. Regular updates will be provided to the supervisory team, with final outcomes presented to the QUT engineering faculty and associated partners.

Table 3.1: Project Stakeholders

Stakeholder	Interest / Role
Prof. Niko Suenderhauf	Project Supervisor. Provides overarching technical guidance, assesses progress, and ensures the academic rigour of the scene change detection implementation.
Dr. Scarlett Raine	Dataset contact and domain expert for underwater research at QCR. Provides insights into the ecological relevance of the project and ensures that the research aligns with real-world coral reef monitoring needs.
QUT Centre for Robotics	High-value stakeholder and end-user. Interested in the integration of the SCD pipeline for advanced robotics mapping and autonomy research.
AIMS	High-value stakeholder and provider of the EcoRRAP dataset. AIMS is interested in utilizing advanced 3D reconstruction to track coral reef health and structural changes.

3.2 Resources

Training and optimizing Gaussian splatting models requires significant computational power and high-quality data. Successful completion of this project requires the following key resources:

- **Dataset:** The AIMS EcoRRAP dataset will serve as the core testing foundation. This high-fidelity dataset contains time-series field photogrammetry of coral reef

clusters, which is absolutely essential for training and validating the change detection pipeline.

- **Computing Hardware:** Training Gaussian Splatting models is a memory and compute-intensive process. The project necessitates access to a high-performance workstation equipped with a top-tier NVIDIA GPU boasting a minimum of 24 GB of VRAM (For example a NVIDIA RTX 3090, RTX 4090, or A-series equivalents) to handle the optimization of millions of 3D primitives without running out of memory.
- **Software & Frameworks:** Python for script development, PyTorch for executing 3DGS optimizations, and specialized 3DGS CUDA kernels. Agisoft Metashape or COLMAP may also be required for initial camera pose estimations (Structure from Motion) prior to splatting.

Project Methodology

This project will follow a structured, four-phase pipeline development methodology tailored for applied computer vision and robotics research. It is designed to be highly adaptable, allowing for the iterative testing and integration of state-of-the-art methods.

4.1 Phase 1: Foundation and Method Selection

The initial phase focuses on establishing the theoretical and data foundations required for the project.

- **Literature & Candidate Selection:** Conduct a deep dive into recent (2024-2026) underwater 3D Gaussian Splatting (3DGS) architectures. Select 2-4 candidate methods (e.g., DualPhys-GS, WaterGS) based on open-source code availability, physical grounding, and geometric robustness.
- **Data Preparation:** Secure and preprocess the AIMS EcoRRAP dataset. This involves data cleaning, establishing camera poses, and performing initial Structure

from Motion (SfM) using tools like COLMAP or Agisoft Metashape to generate the sparse point clouds required to initialize 3DGS.

- **Pipeline Scaffolding:** Set up the high-performance computing environment, ensuring PyTorch, CUDA dependencies, and 3DGS rasterizers are properly configured.

4.2 Phase 2: Baseline Reconstruction and Optimization

Before change detection can occur, accurate baseline models must be generated and geometrically evaluated.

- **Model Training:** Train the 2-4 selected underwater 3DGS models on time-sequential data from the EcoRRAP dataset (e.g., Time A and Time B).
- **Reconstruction Evaluation:** Evaluate the visual synthesis quality using standard metrics (PSNR, SSIM, LPIPS).
- **Geometric Validation & Filtering:** Critically assess the geometric fidelity of the primitives. At the conclusion of this phase, a review will determine if additional processing or geometric filtering is required. For instance, to achieve the millimeter accuracy necessary for detecting fine-scale coral changes, techniques like SuGaR [30] may be implemented to force the 3D Gaussians to align tightly to the actual physical surfaces, thereby suppressing floating artifacts and water caustics.

4.3 Phase 3: Scene Change Detection (SCD) Integration

This phase bridges the gap between static reconstruction and dynamic analysis.

- **Pipeline Integration:** Implement the GS-DIFF (or similar primitive-space) change detection pipeline on top of the trained underwater 3DGS models.
- **Temporal Alignment:** Ensure precise spatial alignment between the models of Time A and Time B. If models are trained independently, implement robust registration techniques to align the point clouds and Gaussians into a shared global coordinate space.

- **Kernel Adaptation:** Fine-tune and adapt the geometric and appearance kernels used by the SCD algorithm to account for the unique optical attenuation and scattering characteristics inherent to the underwater reconstructions.

4.4 Phase 4: Comparative Evaluation and Synthesis

The final phase evaluates the entire pipeline's efficacy and answers the project's core research questions.

- **SCD Efficacy in Coral Scenes:** Evaluate the overall success of the primitive-space change detection pipeline in accurately identifying structural changes in coral reefs while ignoring transient lighting and water shifts.
- **Foundation Comparison:** Compare how the choice of the underlying 3DGS method (from Phase 2) impacts the final change detection accuracy. Identify which underwater adaptation provides the most robust foundation for SCD.
- **Documentation & Recommendations:** Synthesize findings into the final report, providing actionable insights for the QUT Centre for Robotics and AIMS on the viability of this pipeline for long-term ecological monitoring.

Deliverables

The project is expected to yield the following key deliverables upon completion:

1. **Final Project Report:** A comprehensive thesis document detailing the theoretical background, methodological approach, implementation specifics, and evaluation results of the underwater scene change detection pipeline.
2. **Working Code Repository & Pipeline:** A fully operational, documented, and reproducible codebase that executes the selected underwater 3DGS reconstructions and the primitive-space change detection algorithm.

3. **Detailed 3D Gaussian Splat Models:** High-fidelity, water-removed 3D Gaussian Splatting reconstructions of the AIMS EcoRRAP coral reef datasets, serving as the foundational maps.
4. **Dataset of Change Maps:** A comprehensive set of extracted change maps (Geometric, Colour, and Full) demonstrating structural changes and anomalies across the temporal datasets.
5. **Project Presentation:** A final presentation delivered to the supervisory team, stakeholders, and QUT faculty, providing an overview of the pipeline's capabilities and the results achieved on the coral datasets.

Quality & Sustainability

6.1 Quality

The quality of the final deliverable will be validated against a set of performance criteria using standard scene change detection metrics, alongside strict software engineering practices:

- **Accuracy Metrics:** Evaluating Mean Intersection over Union (mIoU) and F1 score against baseline render-then-compare methods.
- **Disambiguation:** The ability to successfully separate true structural changes from surface-level color changes accurately.
- **Robustness:** Performance under the inherent underwater noise, turbidity, and optical attenuation found in the EcoRRAP dataset.
- **Software Engineering Practices:** The codebase will be maintained using Git version control, ensuring a transparent development history. Unit testing will be employed for core algorithmic components (such as kernel computation) to ensure reliability and reproducible research outputs.

6.2 Sustainability

Sustainability will be considered through both computational and environmental lenses:

- **Computational Efficiency:** Ensuring the primitive comparison runs efficiently in 3D space, significantly reducing the carbon footprint of the pipeline by avoiding power-intensive render-then-compare loops across hundreds of synthetic 2D viewpoints.
- **Environmental Impact:** By bridging the gap between raw data collection and actionable spatial conservation planning, this pipeline directly supports marine conservation initiatives (e.g., Global Biodiversity Framework Targets). More accurate and scalable tracking of coral reef structural health enables better-informed ecological management.
- **Knowledge Transfer:** Producing comprehensive documentation and a well-commented codebase to ensure the project's findings can be efficiently built upon in future QUT Centre for Robotics and AIMS research projects, preventing redundant computational work.

Risks, Requirements & Constraints

7.1 Constraints & Requirements

- **Timeline & Compute Factor:** The project is strictly constrained by a two-semester timeline. A major requirement is factoring in the substantial time required to train multiple computationally expensive 3DGS models before the change detection evaluation phase can even begin.
- **Data Availability & Suitability:** The project relies heavily on acquiring suitable temporal data. A critical constraint is finding data that surveys an identical geographical location with a sufficient time gap to demonstrate structural change.

- **Data Security & Compliance:** Any data utilized, particularly the AIMS EcoRAP dataset, must be securely managed and used strictly within the bounds of the established data-sharing or borrowing agreements.
- **Hardware Access:** Training 3DGS models requires continuous access to high-end GPUs. A strict constraint is securing this compute time locally; failure to do so will require pivoting the project budget into cloud computing solutions (e.g., AWS, RunPod) to train the pipeline.

7.2 Risks and Blockers

- **Algorithm Failing on Underwater Noise:** If the inherent underwater noise, scattering, and optical attenuation cannot be adequately filtered during the initial reconstruction phase, these errors will propagate directly into the GS-DIFF model, causing the algorithm to falsely identify noise as structural change.
- **Extreme Environmental Change:** Too much change between timestamps could result in completely uncorrelated reconstructions, making temporal alignment and tracking impossible.
- **Moving Features:** Transient features in the dataset (like marine life, moving sand, or shifting shadows) could duplicate on the change map, heavily cluttering the output and masking actual coral growth or decay.
- **Complexity of Coral:** Coral is highly geometrically complex and can be exceedingly difficult to model accurately via Gaussian splatting, especially when measured under noisy underwater conditions.

Project Timeline

The project is scheduled over two semesters, concluding in October 2027. The progression of the six core phases is visualized in the Gantt chart below, accounting for the substantial time required for algorithm development and model training.

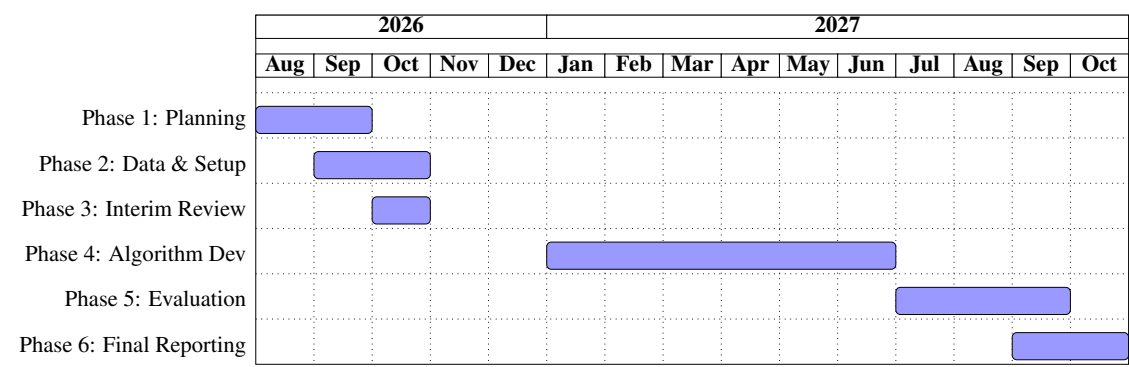


Figure 8.1: Project Timeline and Phase Progression (Aug 2026 – Oct 2027)

Table 8.1: Key Project Milestones

Phase	Timeline	Milestone
Phase 1: Planning	Aug – Sep 2026	Project Proposal (this document).
Phase 2: Data & Setup	Sep – Oct 2026	EcoRRAP dataset secured, baseline models generated.
Phase 3: Interim Review	October 2026	Presentation of progress, initial reconstruction tests.
Phase 4: Algorithm Dev	Feb – Jun 2027	GS-DIFF pipeline integration with underwater adaptations.
Phase 5: Evaluation	Jul – Sep 2027	Extracting change maps and evaluating structural changes.
Phase 6: Final Reporting	Sep – Oct 2027	Full results, Final Report, Oral Presentation.

Management of Project Changes

Any requested changes to the project scope by the supervisor or other stakeholders will be formally managed. A change request will be documented, outlining the nature of the change, its impact on the timeline and resources, and the reason for the change. This document will be reviewed with the supervisor before any new work is undertaken. This scope of work will be updated to reflect any approved changes, and version control will be maintained using Git.

Sign off

By signing below, the Student Engineer agrees to undertake the project as outlined in this Scope of Work.

STUDENT ENGINEER

DATE

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